Near-side azimuthal and pseudo-rapidity correlations of neutral strange baryons and mesons in d+Au and Au+Au collisions at $\sqrt{s_{NN}}$ = 200 GeV

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We present results on near-side di-hadron correlations of neutral strange baryons (Λ,$\bar{\Lambda}$) and mesons (K^0_S) in the intermediate p_T regime (p_T = 2-6 GeV/c) in d+Au and Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV measured by the STAR experiment at RHIC. The study is performed for both identified strange trigger particles associated with charged particles as well as charged trigger particles using strange associated particles. The di-hadron distribution in ($\Delta\phi$, $\Delta\eta$) space is decomposed into a narrow jet-like peak at small angular separation $\Delta\phi$ and a component which is narrow in $\Delta\phi$ and extended in $\Delta\eta$, the "ridge". We study the near-side yield of the ridge and jet-like components as a function of centrality of the collision and transverse momentum (p_T) of trigger and associated particles to look for possible flavor and baryon/meson differences. The magnitude, p_T spectra and particle composition of the jet-like component are similar to those in d+Au collisions. The strength of the ridge increases steeply with centrality in Au+Au collisions for all studied particle species. The p_T spectra and baryon/meson composition of the ridge resemble those for the bulk particle production. The results on the ridge are discussed with theoretical models.

PACS numbers: 25.75.-qRelativistic heavy-ion collisions 25.75.GzParticle correlations and fluctuations

I. INTRODUCTION

Ultra-relativistic heavy-ion collisions create a unique environment for the investigation of nuclear matter at extreme temperatures and energy densities in the laboratory. Already the first measurements of nuclear modification factors [1, 2] and anisotropic elliptic flow at the Relativistic Heavy Ion Collider (RHIC) at $\sqrt{s_{NN}} = 200$ GeV showed that the nuclear medium created is opaque to particles with large transverse momentum (p_T), is described by partonic degrees of freedom and has properties close to those expected of a perfect fluid.

As compared to nuclear modification factors, studies of jets and their modification in the medium provide a better sensitivity to the medium properties [3]. Azimuthal correlations of particles with large p_T can be used to study processes associated with jets. Previous investigations of two-particle azimuthal correlations in central Au+Au collisions span a large p_T range (p_T = 0.15 - 15 GeV/c) resulted in several interesting observations. There are striking differences between those correlations in p+p and d+Au and those in central Au+Au collisions: a) the disappearance of the away-side jet at intermediate p_T = 2-6 GeV/c consistent with partonic energy loss in medium [4], b) the increase of the number of low p_T (p_T = 0.15-2 GeV/c) particles on the away-side from the trigger particle with concomitant shape modifications suggesting conical particle emission [5], c) *Punch through* or the ability of away-side jets, with sufficiently large p_T ($p_T > 8$ GeV/c), to traverse the hot and dense medium [6], and d) the presence of a correlation on the near-side extended in pseudo-rapidity ($\Delta\eta$), commonly referred to as the *ridge* at intermediate p_T [7, 8].

It is interesting to note that besides the shape modifications of jet-like correlations and the existence of the ridge in the intermediate transverse momentum range, the production mechanism of hadrons seems to differ from simple fragmentation. Baryon production is less suppressed in central Au+Au collisions than that of mesons resulting in baryon/meson ratios enhanced in central Au+Au collisions relative to p+p and d+Au collisions [9, 10]. The measured baryon/meson ratios increase with p_T until reaching their maximum value of ≈ 3 relative to p+p colliisions at $p_T \approx 3$ GeV/c in both the strange and non-strange quark sectors. A fall-off of the baryon/meson ratio is observed for $p_T > 3$ GeV/c and both the strange and non-strange baryon/meson ratios in Au+Au collisions approach the values measured in p+p collisions at $p_T \approx 6$ GeV/c.

This shows that in the intermediate p_T range at RHIC energies unmodified jet fragmentation is not the dominant source of particle production. Parton recombination and coalescence models are an alternative particle production mechanism [11–15]. In these models, competition between recombination and fragmentation results in a shift in the onset of the perturbative regime to higher transverse momenta, $p_T \approx$ 4-6 GeV/c. Due to the steeply falling transverse momentum spectrum of partons, fragmentation is a much less efficient particle production mechanism than recombination. Moreover, the steeply falling parton spectrum favors the recombination of three quarks to form a baryon over the recombination of two quarks to form a meson with the same p_T. Such mechanisms naturally lead to increased production of baryons relative to mesons, in qualitative agreement with the data.

In this paper, studies of two-particle correlations on the near-side in d+Au, Cu+Cu and Au+Au collisions at $\sqrt{s_{NN}}$ = 200 GeV measured by the STAR experiment are presented. Results from two-particle correlations in pseudo-rapidity and azimuth for neutral strange baryons (Λ, $\bar{\Lambda}$) and mesons (K^0_S) at intermediate p_T (2<p_T <6

GeV/c) in the different collision systems are compared to unidentified hadron correlations. Both identified trigger particles associated with charged particles (K^0_S-h, $\Lambda,\bar{\Lambda}$-h) and charged trigger particles associated with identified strange particles (h-K^0_S, h-$\Lambda,\bar{\Lambda}$) are studied. The near-side jet-like yield is studied as a function of centrality of the collision and transverse momentum of trigger and associated particles to look for possible flavor, baryon/meson and particle/antiparticle differences. The composition of the jet-like correlation is studied using identified associated particles to investigate possible production mechanisms. Results are compared to expectations from PYTHIA.

II. EXPERIMENTAL SETUP AND PARTICLE RECONSTRUCTION

The results presented in this paper are based on data measured by the STAR experiment from d+Au collisions at $\sqrt{s_{NN}}$ = 200 GeV in 2003, Cu+Cu collisions at 200 GeV in 2005, and Au+Au collisions at 200 GeV in 2004. The d+Au events were selected using a minimally biased (MB) trigger requiring at least one beam-rapidity neutron in the Zero Degree Calorimeter (ZDC), located at 18 m from the nominal interaction point in the Au beam direction, accepting 95±3% of the Au+Au hadronic cross section [16]. For Cu+Cu collisions, the MB trigger was based on the combined signals from the Beam-Beam Counters (BBC) at forward rapidity ($3.3 < |\eta| < 5.0$) and a coincidence between the ZDCs. The MB trigger for Au+Au collisions was obtained using a ZDC coincidence, a signal in both BBCs and a minimum charged particle multiplicity in an array of scintillator slats arranged in a barrel, the Central Trigger Barrel (CTB), to reject non-hadronic interactions.

For Au+Au collisions at $\sqrt{s_{NN}}$ = 200 GeV, an additional online trigger for central collisions was used. This trigger was based on the energy deposited in the ZDCs in combination with the multiplicity in the CTB. The central trigger sampled the most central 12% of the total hadronic cross section.

In order to achieve a more uniform detector acceptance, only those events with the primary vertex position along the longitudinal beam direction (z) within 30 cm of the center of the STAR detector were used for the analysis. For the d+Au collisions this was expanded to $|z| < 50$ cm. The number of events after the vertex cut in individual data samples is summarized in table II.

TABLE I: Number of events after cuts (see text) in the data samples analyzed.

System	Centrality	No. of events [10^6]
d+Au	0-95%	3
Cu+Cu	0-60%	38
Au+Au	0-80%	28
Au+Au	0-12%	17

STAR [17] is a large acceptance, multi-purpose spectrometer consisting of several detectors inside a large solenoidal magnet with a uniform magnetic field of 0.5 T applied along the beam line. This analysis is based exclusively on charged particle tracks detected and reconstructed in the Time Projection Chamber (TPC) [18]. The TPC has a pseudo-rapidity acceptance $|\eta| < 1.5$ and full azimuthal coverage. The TPC has 45 pad rows in the radial direction allowing up to 45 independent spatial and energy loss (dE/dx) measurements along each charged particle track. The momentum resolution is determined to be $\Delta k/k \sim 0.0078 + 0.0098 \cdot p_T$ (GeV/c) [18].

III. METHOD

A. Correlation technique

The analysis in this paper follows the method in [19]. Tracks used in this analysis were required to have at least 15 fit points in the TPC, a DCA to the primary vertex of less than 1 cm, and a pseudorapidity $|\eta| < 1.0$. A high-p_T trigger particle was selected and the raw distribution of associated tracks relative to that trigger in pseudorapidity ($\Delta\eta$) and azimuth ($\Delta\phi$) is formed. This distribution, $d^2N_{\rm raw}/d\Delta\phi\, d\Delta\eta$, is normalized by the number of trigger particles, $N_{\rm trigger}$, and corrected for the efficiency and acceptance of associated tracks ε:

$$\frac{d^2N}{d\Delta\phi\, d\Delta\eta}(\Delta\phi,\Delta\eta) = \frac{1}{N_{\rm trigger}}\frac{d^2N_{\rm raw}}{d\Delta\phi d\Delta\eta} \frac{1}{\varepsilon_{\rm assoc}(\phi,\eta)}\frac{1}{\varepsilon_{\rm pair}(\Delta\phi,\Delta\eta)}. \quad (1)$$

The efficiency correction $\varepsilon_{\rm assoc}(\phi,\eta)$ is a correction for the TPC single particle reconstruction efficiency and $\varepsilon_{\rm pair}(\Delta\phi,\Delta\eta)$ is a correction for the finite TPC track-pair acceptance in $\Delta\phi$ and $\Delta\eta$. Since the correlations are normalized per trigger, the efficiency correction is only applied for the associated particle. The data presented in this paper are averaged between positive and negative $\Delta\phi$ and $\Delta\eta$ regions and are reflected about $\Delta\phi$ =0 and $\Delta\eta$ =0 in the plots.

B. Single charged track efficiency correction

For identified trigger particles, the efficiency correction is the correction for unidentified charged hadrons, identical to the efficiency correction applied in [19]. The single particle reconstruction efficiency in the TPC is determined by simulating the detector response to a particle and embedding these signals into a real event. The efficiency for detecting a single track as a function of p_T, η, and centrality is determined from the number of simulated particles which were successfully reconstructed. The single track efficiency is approximately constant for

$p_T > 2$ GeV/c and ranges from around 75% for central Au+Au events to around 85% for peripheral Cu+Cu events. The efficiency for reconstructing a track in d+Au is 89%. The reconstruction efficiency for Λ, $\bar{\Lambda}$, and K^0_S reconstruction is determined in a similar way, but forcing the simulated particle to decay through the channel in equation III F. The efficiency for Λ, $\bar{\Lambda}$, and K^0_S ranges from 8% to 15%, increasing with momentum and decreasing with system size.

The systematic uncertainty on the efficiency correction, 5%, is strongly correlated across centralities and p_T bins for each data set but not between data sets. In the correlations, each track pair is corrected for the efficiency for reconstructing the associated particle. Since the correlations are normalized by the number of trigger particles, no correction for the efficiency of the trigger particle is necessary.

C. Pair acceptance correction

With the restriction that each track falls within $|\eta| <$ 1.0, there is a limited acceptance for track pairs. For $\Delta\eta \approx 0$, the geometric acceptance of the TPC for track pairs is $\approx$ 100%, however, near $\Delta\eta \approx 2$ the acceptance is close to 0%. In azimuth the acceptance is limited by the 12 TPC sector boundaries, leading to dips in the acceptance of track pairs in azimuth. To correct for the geometric acceptance, a mixed event analysis was done using trigger particles from one event combined with associated particles from another event, as done in [20]. The position of the event vertices were required to be within 2 cm of each other along the beam axis and have the same charged particle multiplicity within 10 particles. Each event with a trigger particle was mixed with ten other events. Christine: double check details.

D. Track merging correction

The track merging effect in unidentified hadron (h) correlations discussed in [19] is also present for V^0-h and h-V^0 correlations. This effect leads to a dip at small $\Delta\phi$ and $\Delta\eta$ due to overlap between the trigger and associated particles. When one of the particles is a V^0, this overlap is between one of the V^0 daughter particles and the unidentified hadron. The size of the dip due to track merging depends strongly on the relative momenta of the particle pair. This effect is larger when the overlapping particles' momenta are closer. For V^0-h correlations, the typical associated particle momentum is approximately 1.5 GeV/c. Since the K^0_S decay is symmetric, this means that the track merging effect is greatest for K^0_S-h correlations with a trigger K^0_S momentum of approximately 3 GeV/c. In a Λ decay, the proton daughter carries more of the Λ momentum than the pion daughter. Therefore this effect is larger for lower momentum Λ trigger particles. Since track merging affects both signal and background

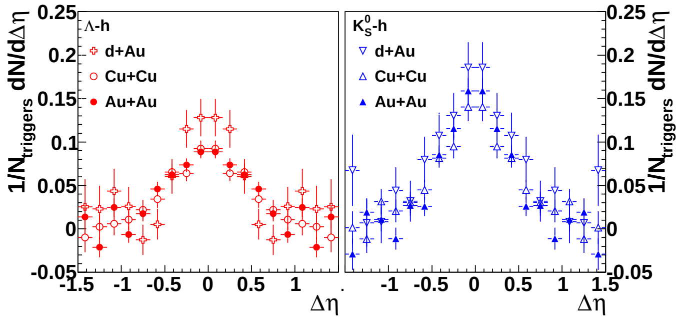

FIG. 1: $\frac{dN_J}{d\Delta\eta}$ for Λ, $\bar{\Lambda}$-h (left) and K^0_S-h (right) correlations for minimum bias d+Au collisions, 0-20% Cu+Cu collisions, and 40-80% Au+Au correlations. The data have been reflected about $\Delta\eta = 0$.

particles and the signal sits on top of a large combinatorial background, the effect is larger for collisions with a higher charged track multiplicity. Since the dip in V^0-h and h-V^0 correlations is the result of a V^0 daughter merged with an unidentified hadron, the dip is wider in $\Delta\phi$ and $\Delta\eta$ than in unidentified hadron correlations. When the track merging dip is present, it is corrected by fitting a Gaussian to the peak, excluding the region impacted by track merging, and using the fit to extract the yield.

E. Yield extraction

Extraction of the yield in this paper follows the method and the notation in [19, 20]. The jet-like correlation is narrow in both $\Delta\phi$ and $\Delta\eta$ and is contained within the kinematic cuts in $p_T^{\rm trigger}$ and $p_T^{\rm associated}$ used in this analysis. The di-hadron correlation from equation 1 is projected onto the $\Delta\eta$ axis:

$$\left.\frac{dN}{d\Delta\eta}\right|_{a,b} \equiv \int_a^b d\Delta\phi \frac{d^2N}{d\Delta\phi d\Delta\eta}. \qquad (2)$$

All other correlations, including those from v_2, $V_{3\Delta}$, and higher order flow harmonics, are assumed to be independent of $\Delta\eta$ within the η acceptance of the analysis, consistent with [20–23]. This means that the jet-like correlation can be determined by:

$$\frac{dN_J}{d\Delta\eta}(\Delta\eta) = \left.\frac{dN}{d\Delta\eta}\right|_{-0.78,0.78} - b_{\Delta\eta} \qquad (3)$$

where $b_{\Delta\eta}$ is a constant offset determined by fitting a constant background $b_{\Delta\eta}$ plus a Gaussian to $\frac{dN_J}{d\Delta\eta}(\Delta\eta)$. Sample correlations are given in Figure 1. When the dip due to track merging is negligible, the yield determined from fit is discarded to avoid any assumptions about the shape of the peak and instead we integrate equation 3 over $\Delta\eta$ using bin counting to determine the jet-like yield $Y_J^{\Delta\eta}$:

$$Y_J^{\Delta\eta} = \int_{-0.78}^{0.78} d\Delta\eta \, \frac{dN_J}{d\Delta\eta}(\Delta\eta) \; . \tag{4}$$

When it is not possible, the yield from the Gaussian fit is used, excluding the region affected by track merging, and the statistical errors from the fit exceed any error from the assumption of a Gaussian shape.

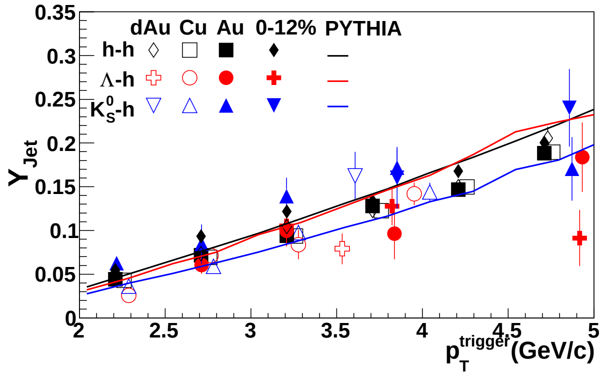

FIG. 2: The jet-like yield as a function of $p_T^{\rm trigger}$ for h-h [19], K_S^0-h, and $\Lambda,\bar{\Lambda}$-h correlations in d+Au, Cu+Cu, and Au+Au collisions at $\sqrt{s_{NN}}$ = 200 GeV. The data from Au+Au collisions are shown in two centrality bins, 40-80% and 0-12%. The data are compared to PYTHIA [25] calculations using the Perugia 2011 tune [26].

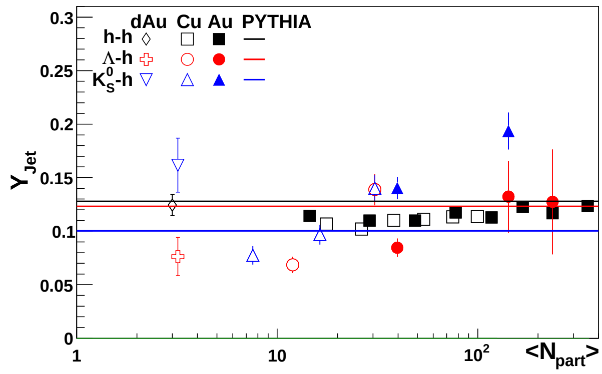

FIG. 3: $N_{\rm part}$ dependence of the jet-like yield of h-h, K_S^0-h, and $\Lambda,\bar{\Lambda}$-h correlations in d+Au, Cu+Cu, and Au+Au collisions at $\sqrt{s_{NN}}$ = 200 GeV. The data from Au+Au collisions are shown in two centrality bins, 40-80% and 0-12%. The data are compared to PYTHIA [25] calculations using the Perugia 2011 tune [26].

F. Particle identification

We identify weakly decaying neutral strange (V^0) particles Λ, $\bar{\Lambda}$ and K_S^0 by topological reconstruction of their decay vertices from their charged hadron daughters measured in the TPC as described in [24]:

$$\begin{aligned} \Lambda &\rightarrow p+\pi^-, BR = (63.9 \pm 0.5)\% \\ \bar{\Lambda} &\rightarrow \bar{p}+\pi^+, BR = (63.9 \pm 0.5)\% \\ K_S^0 &\rightarrow \pi^+ + \pi^-, BR = (68.95 \pm 0.14)\% \end{aligned} \tag{5}$$

where BR denotes the branching ratio. The STAR V^0 finding reconstruction software pairs oppositely charged particle tracks into V^0 candidates. Cuts are optimized for each data set and yield signal to background ratio of at $\geq$ 15:1. For the analyses presented here, no significant difference was observed between results with Λ and $\bar{\Lambda}$ trigger and associated particles so the correlations were added to increase statistics.

IV. RESULTS

A. Correlations with identified strange trigger particles

The jet-like yield as a function of $p_T^{\rm trigger}$ is shown in figure 2 for h-h [19], K_S^0-h, and $\Lambda,\bar{\Lambda}$-h correlations for d+Au, Cu+Cu, and Au+Au collisions at $\sqrt{s_{NN}}$ = 200 GeV. Due to residual track merging effects, fits are used for $\Lambda,\bar{\Lambda}$-h correlations in Cu+Cu collisions with 2.0 < $p_T^{\rm trigger}$ < 3.0 GeV/c, in 0-12% Au+Au collisions for 3.0 < $p_T^{\rm trigger}$ < 4.5 GeV/c, and in 40-80% Au+Au collisions for 2.0 < $p_T^{\rm trigger}$ < 4.5 GeV/c and for K_S^0-h correlations in 0-12% Au+Au collisions for 3.0 < $p_T^{\rm trigger}$ < 3.5 GeV/c and 40-80% Au+Au collisions for 2.0 < $p_T^{\rm trigger}$ < 4.5 GeV/c. As observed for h-h correlations in [19], there is no significant difference in the yields between the collision systems. This includes no significant difference between results from Au+Au collisions in 40-80% and 0-12% central collisions, although there is a hint that the yield may be lower in central collisions. The jet-like yield from K_S^0-h correlations is systematically above the jet-like yield from h-h correlations and the jet-like yield from $\Lambda,\bar{\Lambda}$-h is systematically below the jet-like yield from h-h correlations, particularly in Au+Au collisions. Data are compared to PYTHIA calculations from the Perugia 2011 [26] tune. PYTHIA describes the trend of the jet-like yield well for all particle species. However, the trigger particle type ordering in PYTHIA is opposite that observed in the data, with the $\Lambda,\bar{\Lambda}$-h jet-like yield higher than the h-h jet-like yield, which is higher than K_S^0-h jet-like yield. Similar mass ordering is observed in the Perugia 2010 tune.

The jet-like yield as a function of $N_{\rm part}$ is shown in figure 3 for h-h [19], K_S^0-h, and $\Lambda,\bar{\Lambda}$-h correlations for d+Au, Cu+Cu, and Au+Au collisions at $\sqrt{s_{NN}}$ = 200 GeV. All yields are determined using bin counting. While there is no centrality dependence in the jet-like yield of h-h correlations, there is an increase in the yield with centrality for K_S^0-h correlations and a hint that there may be a similar increase in $\Lambda,\bar{\Lambda}$-h correlations. This

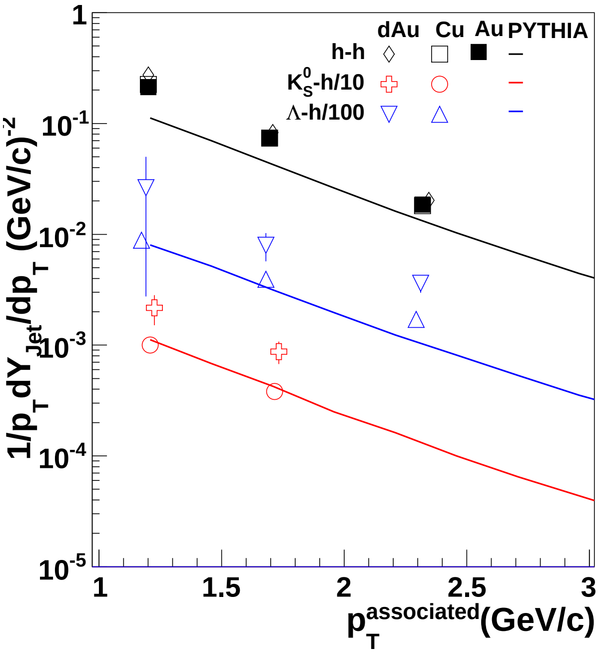

FIG. 4: The jet-like yield as a function of $p_T^{\text{associated}}$ for h-h [19], K_S^0-h, and $\Lambda,\bar{\Lambda}$-h correlations in d+Au, Cu+Cu, and Au+Au collsions at $\sqrt{s_{NN}}$ = 200 GeV. The data are compared to PYTHIA [25] calculations using the Perugia 2011 tune [26].

could be caused by many different factors. If strange jets experience more energy loss than light flavored jets, the jet-like yield associated with a strange trigger particle could increase even as the jet-like yield stays the same for light particles. If lower energy strange jets are suppressed more than lower energy light jets, the surviving spectrum of jets could be skewed towards higher energy jets and the corresponding jet-like yield would be higher. An increased yield could also come from a modification of fragmentation functions inside the medium, either from an enhancement in the production of lower energy particles in particles containing a strange jet or from an enhancement in the production of higher energy strange particles in light jets. Jana please review the preceeding logic. Jana I also removed the fits to the different particle species since I don't think it'd be very meaningful These data are compared to PYTHIA [25] calculations from the Perugia 2011 [26] tune. PYTHIA gets the trigger particle type ordering wrong, even for d+Au collisions, indicating that the production mechanism for strange particles in PYTHIA may need to be refined.

The jet-like yield as a function of $p_T^{\text{associated}}$ is shown in figure 3 for h-h [19], K_S^0-h, and $\Lambda,\bar{\Lambda}$-h correlations for d+Au, Cu+Cu, and Au+Au collisions at $\sqrt{s_{NN}}$ = 200 GeV. All yields are determined using bin counting. The $\Lambda,\bar{\Lambda}$-h and K_S^0-h correlations are only shown for d+Au and Cu+Cu since residual track merging made measurements in Au+Au collisions difficult. These data are compared to PYTHIA [25] calculations from the Perugia 2011 [26] tune. PYTHIA describes the Cu+Cu data better than the d+Au data. Jana, I removed the fits and the PID'd Au+Au points. The Au+Au points all looked kind of iffy - I wondered whether we could really extract a yield. They either had large errors or the background was maybe not flat. I've left all of the plots in the analysis note for now. Overall I think that this is an issue we should discuss.

B. Charged-V^0 correlations

It is possible to study the composition of the jet-like correlation using identified associated particles. The statistics of the d+Au data were limited and the Au+Au data set was limited by the presence of residual track merging. Therefore it was only possible to extract data from Cu+Cu collisions. Since the jet-like correlation in h-h correlations has been shown to be consistent across collision systems [19, 20], the composition of the jet-like correlation is expected to be the same for all collision systems. These data are shown in figure 5 and compared to data from inclusive spectra in p+p collisions from STARadd reference and ALICEadd reference and from PYTHIA [25] calculations from the Perugia 2010 and 2011 [26] tunes. The inclusive ratios are consistent for both tunes so only one line is shown. The Cu+Cu data are consistent with the inclusive particle ratios from p+p, which is expected if the dominant production mechanism for strange particles at high momenta is mainly fragmentation. The composition of the jet-like yield in PYTHIA are approximately two standard deviations away from the value measured in Cu+Cu collisions. This indicates that PYTHIA may be underestimating the fraction of strange particles produced from the fragmentation of jets, which could be an explanation for the difficulties PYTHIA has reproducing strange particle spectra.

V. DISCUSSION

VI. CONCLUSIONS

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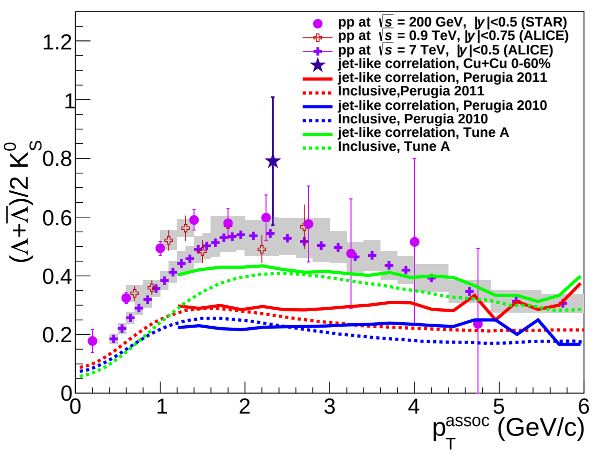

FIG. 5: Λ/K^0_S ratio measured in inclusive p_T distributions, near-side jet-like and ridge-like correlation peaks in Au+Au collisions together with this ratio obtained from inclusive p_T spectra in p+p collisions.

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